The Critical Line
Why $\mathrm{Re}(s) = 1/2$

Diegо Rincón
Independent

Abstract

The Riemann Hypothesis asserts that all nontrivial zeros of $\zeta(s)$ lie on the critical line $\mathrm{Re}(s) = 1/2$. This paper asks: why this line? The functional equation forces zeros to be symmetric about $\mathrm{Re}(s) = 1/2$; the line is the unique axis of symmetry of the nontrivial zeros. Independently: the Källén–Lehmann spectral weight, the Beurling–Nyman Gram matrix symbol $\sigma(t) = |\zeta(1/2+it)|^2/(1/4+t^2)$, and the Connes absorption spectrum all concentrate their information on this line. What is known: Hardy proved infinitely many zeros are on the critical line (1914); Levinson proved at least 1/3 of nontrivial zeros are on it; Conrey improved to 2/5 (2/5 of zeros on the line, unconditionally). The moments of ζ on the critical line encode the random matrix statistics. The Lindelöf Hypothesis (a consequence of RH) asserts $\zeta(1/2 + it) = O(t^\varepsilon)$. The line is not a conjecture; it is the axis.

1 The Functional Equation Forces the Line

Theorem 1 (Functional equation [1])**.** *Define* $\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$*. Then* $\xi(s) = \xi(1-s)$*, and* ξ *is entire. The nontrivial zeros of* ζ *are the zeros of* ξ*, and they are symmetric under* $s \mapsto 1 - s$*: if* $\zeta(\rho) = 0$ *then* $\zeta(1 - \rho) = 0$*. Combined with complex conjugation (*$\zeta(\bar{\rho}) = 0$*): zeros come in quadruples* $\{\rho, 1-\rho, \bar{\rho}, 1-\bar{\rho}\}$*, collapsing to a pair on* $\mathrm{Re}(s) = 1/2$ *iff* $\rho = 1 - \bar{\rho}$*, i.e.,* $\mathrm{Re}(\rho) = 1/2$*.*

Remark 2 (The critical line as axis of symmetry)*.* The functional equation is a reflection through $\mathrm{Re}(s) = 1/2$. The critical line is the unique fixed set of this reflection. A zero off the line exists in a quadruple; a zero on the line exists in a pair. RH says: all zeros are fixed points of the reflection. No zero breaks the symmetry.

2 What Is Known about Zeros on the Line

Theorem 3 (Hardy [5])**.** *Infinitely many zeros of* $\zeta(s)$ *lie on the critical line* $\mathrm{Re}(s) = 1/2$*.*

Theorem 4 (Levinson [6])**.** *At least* $1/3$ *of the nontrivial zeros of* $\zeta(s)$ *lie on the critical line.*

Theorem 5 (Conrey [7])**.** *At least* $2/5$ *of the nontrivial zeros of* $\zeta(s)$ *lie on the critical line.*

Remark 6 (Numerical evidence). Odlyzko and others have verified that the first 10^{13} zeros (ordered by imaginary part) all lie on the critical line. The proportion of zeros on the line, among those verified, is 1. No zero off the critical line has ever been found.

Remark 7 (Why 2/5 is not enough). The Levinson–Conrey method gives a lower bound on the proportion of zeros on the line. Even if 99% of zeros are on the line, a single zero off it would disprove RH. The approach of counting zeros on the line cannot prove RH; it can only accumulate evidence. A proof must rule out off-line zeros entirely. This is the BN wall, the Connes wall, the Hadamard wall.

3 The Moments of Zeta on the Critical Line

Definition 8 (Moments)**.** The $2k$-th moment of ζ on the critical line: $M_{2k}(T) = \int_0^T |\zeta(1/2+it)|^{2k}\, dt$.

Theorem 9 (Moment asymptotics [8])**.** *1.* $M_2(T) \sim T \log T$ *(Hardy–Littlewood, 1918).*

2. $M_4(T) \sim \frac{1}{2\pi^2} T \log^4 T$ *(Ingham, 1926).*

3. $M_{2k}(T) \sim c_k T (\log T)^{k^2}$ *(conjectured for all k; proved for $k = 1, 2$ unconditionally).*

4. The constant c_k is predicted by random matrix theory (GUE): $c_k = g(k) \cdot a(k)$ where $g(k)$ is a GUE factor and $a(k) = \prod_p a_p(k)$ is an arithmetic factor.

Remark 10 (What moments say). The growth rate $T(\log T)^{k^2}$ is exactly the rate for the characteristic polynomial of a random unitary matrix (GUE) in the limit of large matrix size. This is Montgomery's pair correlation in integrated form: the zeros of ζ on the critical line behave, in distribution, like eigenvalues of random unitary matrices. The moments are a global average encoding the same structure as the local pair correlation statistics.

4 The Lindelöf Hypothesis

Conjecture 11 (Lindelöf Hypothesis [9])**.** *For every $\varepsilon > 0$: $\zeta(1/2+it) = O(|t|^{\varepsilon})$ as $|t| \to \infty$.*

Theorem 12 (Lindelöf follows from RH)**.** *RH implies the Lindelöf Hypothesis.*

Proof sketch. Under RH, $\zeta(s) = O(|s|^{\varepsilon})$ in the strip $1/2 \le \mathrm{Re}(s) \le 1$ by the Phragmén–Lindelöf principle applied between the known bounds $\zeta(1+it) = O(\log t)$ and $\zeta(1/2+it) = O(t^{1/4+\varepsilon})$ (the latter unconditional; RH improves it to $O(t^{\varepsilon})$). $\square$

Remark 13 (Lindelöf and the critical line). The Lindelöf Hypothesis says: ζ grows slowly on the critical line. Zeros close to the line (with $\mathrm{Re}(\rho)$ near 1/2 but not equal) would cause ζ to grow faster. Lindelöf is the growth-rate consequence of RH. It has not been proved unconditionally. The best known unconditional bound: $\zeta(1/2+it) = O(t^{13/84})$ (Huxley, 2002).

5 The Critical Line in BN and Connes

Theorem 14 (The line in BN [1, 2]). *The Gram matrix symbol $\sigma(t) = |\zeta(1/2+it)|^2/(1/4+t^2)$ is evaluated precisely on the critical line $\mathrm{Re}(s) = 1/2$. The Toeplitz approximation $T_N(\sigma)$ uses only the values of ζ on this line. The BN distance d_N converges to zero iff the zeros are confined to the line where $\sigma(t)$ is evaluated. The critical line is not a choice; it is where the symbol lives.*

Theorem 15 (The line in Connes [3, 4]). *In the Connes adèle class space, the scaling operator D has absorption spectrum at the imaginary parts γ of the nontrivial zeros $\rho = \sigma + i\gamma$. Under RH, all $\sigma = 1/2$: the absorption spectrum lies on the imaginary axis. The adèle class space is constructed so that its natural geometry singles out the critical line as the "real axis" of the spectral problem. Specifically: the L^2-norm on the adèle class space is invariant under the symmetry $s \mapsto 1 - s$, which fixes the critical line.*

Remark 16 (The line is the fixed set of the only available symmetry). The functional equation gives the only known global symmetry of ζ. Its fixed set is the critical line. Both BN and Connes are built so that the critical line is the axis of their geometry. The zeros cluster around this axis numerically. RH says the clustering is perfect: no zero deviates. We cannot prove the zeros are fixed by the symmetry even though we know the symmetry exists and the zeros appear to respect it.

6 State of the Art

Theorem 17 (What is known about the critical line). *The following hold unconditionally:*

1. *Infinitely many zeros on the critical line (Hardy, 1914).*
2. *At least $2/5$ of all zeros on the critical line (Conrey, 1989).*
3. *The first 10^{13} zeros are on the critical line (numerical).*
4. $M_2(T) \sim T \log T$, $M_4(T) \sim T \log^4 T/(2\pi^2)$.
5. $\zeta(1/2 + it) = O(t^{13/84+\varepsilon})$ *(Huxley).*
6. *The functional equation reflects zeros through the critical line.*

The following are equivalent to RH (not proved):

1. *All nontrivial zeros have $\mathrm{Re}(\rho) = 1/2$.*
2. $d_\infty^2 = 0$ *(BN permanent residual).*
3. *The absorption spectrum of D lies on the imaginary axis (Connes).*
4. $\zeta(1/2 + it) = O(t^{\varepsilon})$ *(Lindelöf, consequence).*

Remark 18 (The line). The critical line is the most studied object in analytic number theory. Every tool has been aimed at it. Hardy counted zeros on it; Levinson and Conrey measured their density; Odlyzko computed them to high precision; GUE statistics describe their spacing. And yet: the proof that all zeros are on it remains out of reach. The line is not mysterious. It is the axis of the only symmetry the problem has. It is where everything happens. We just cannot prove it.

References

[1] Rincón, D. (2026). The Riemann Hypothesis: A Synthesis. Preprint.

[2] Rincón, D. (2026). The Gram Matrix. Preprint.

[3] Rincón, D. (2026). Connes: Noncommutative Geometric Approach to RH. Preprint.

[4] Rincón, D. (2026). Two Approaches to the Riemann Hypothesis. Preprint.

[5] Hardy, G. H. (1914). Sur les zéros de la fonction $\zeta(s)$ de Riemann. *Comptes Rendus de l'Académie des Sciences*, 158, 1012–1014.

[6] Levinson, N. (1974). More than one third of zeros of Riemann's zeta-function are on $\sigma = 1/2$. *Advances in Mathematics*, 13(4), 383–436.

[7] Conrey, J. B. (1989). More than two fifths of the zeros of the Riemann zeta function are on the critical line. *Journal für die reine und angewandte Mathematik*, 399, 1–26.

[8] Conrey, J. B., Rubinstein, M. O., and Snaith, N. C. (2005). Moments of the derivative of characteristic polynomials. *Communications in Mathematical Physics*, 259(3), 543–569.

[9] Titchmarsh, E. C. (1986). *The Theory of the Riemann Zeta-Function*. 2nd ed. Oxford University Press.